# EMPLOYEE SALARY PREDICTION USING LINEAR REGRESSION

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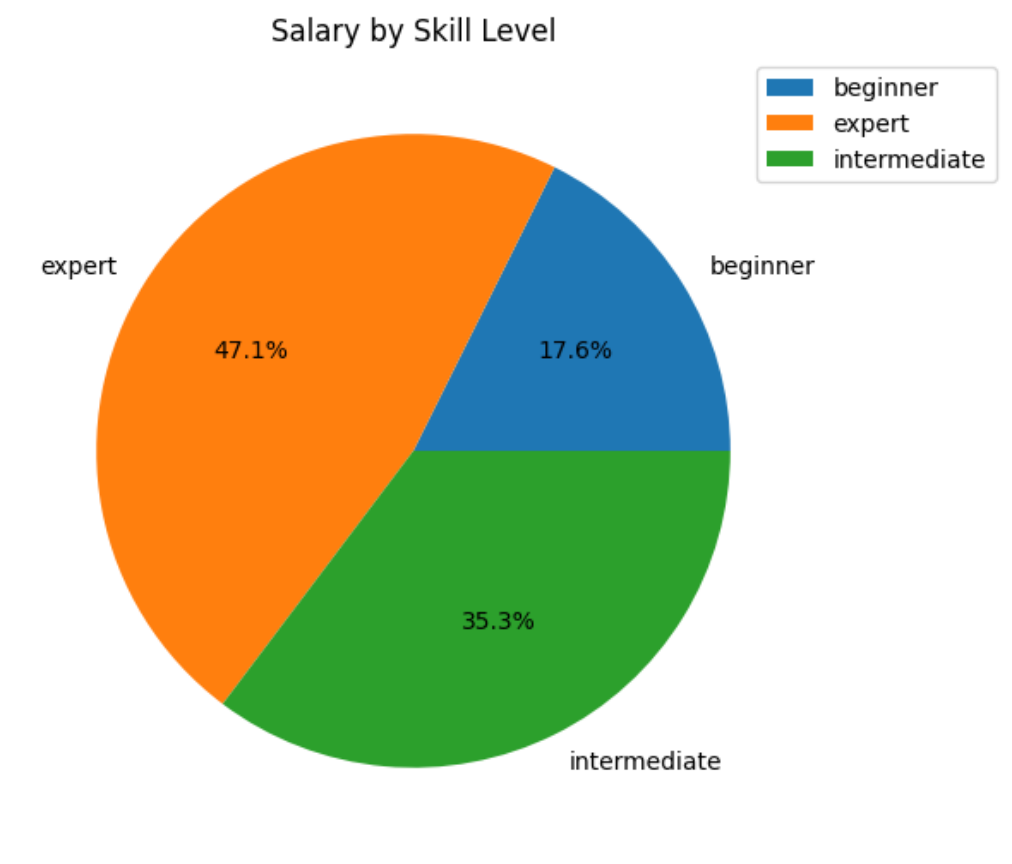
DATA SCIENCE PROJECT

# PROJECT OVERVIEW

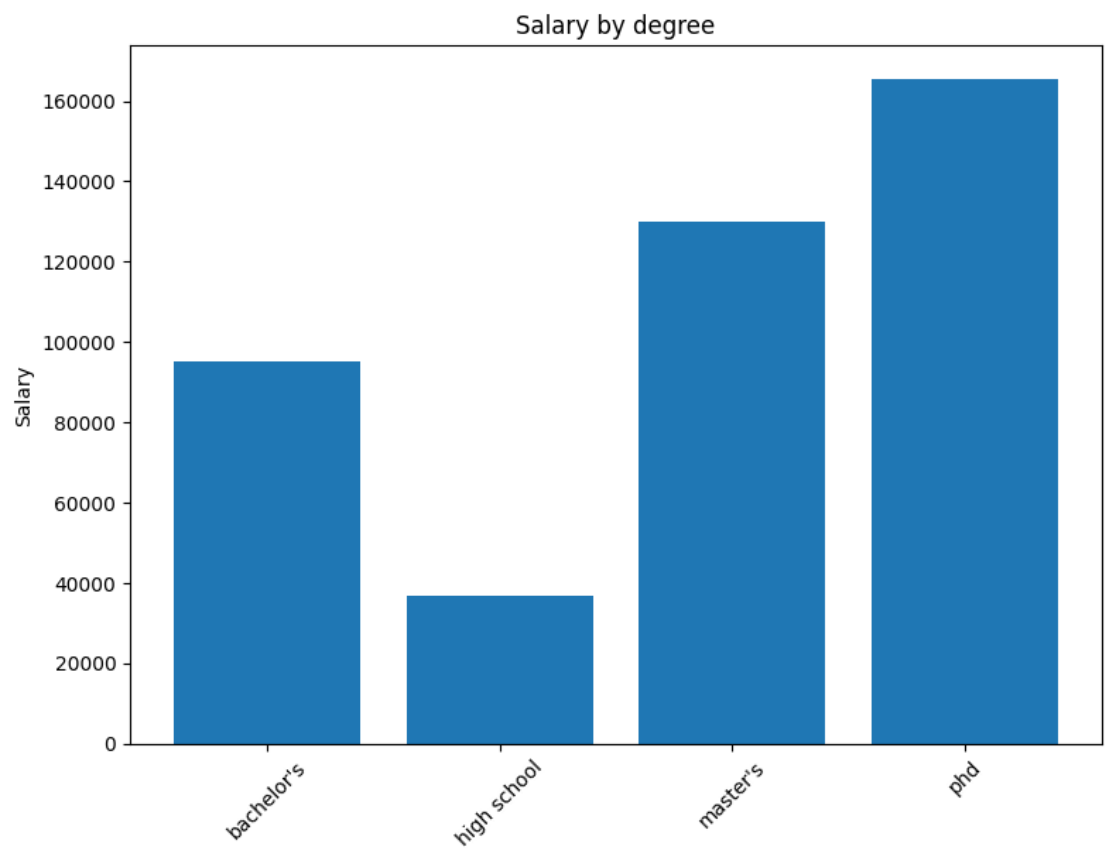
- Goal: Predict employee salaries using machine learning.
- Algorithm: Linear Regression
- Features: Years of Experience, Education Level
- Tools: Python, Pandas, NumPy, Matplotlib, Scikit-learn

# SALARY BY SKILL LEVEL




- This pie chart explains about salary changes according to skill level
- An employee with expert qualification earns 12% more salary than an intermediate..
- This shows experience matter and skill level matters more salary decision making.

# SALARY BY DEGREE




- This bar plot shows comparison between different degrees .
- An employee with Phd has avg salary of 1,60,000 in a company.
- With this plot we can say degree play a major role in salary variation

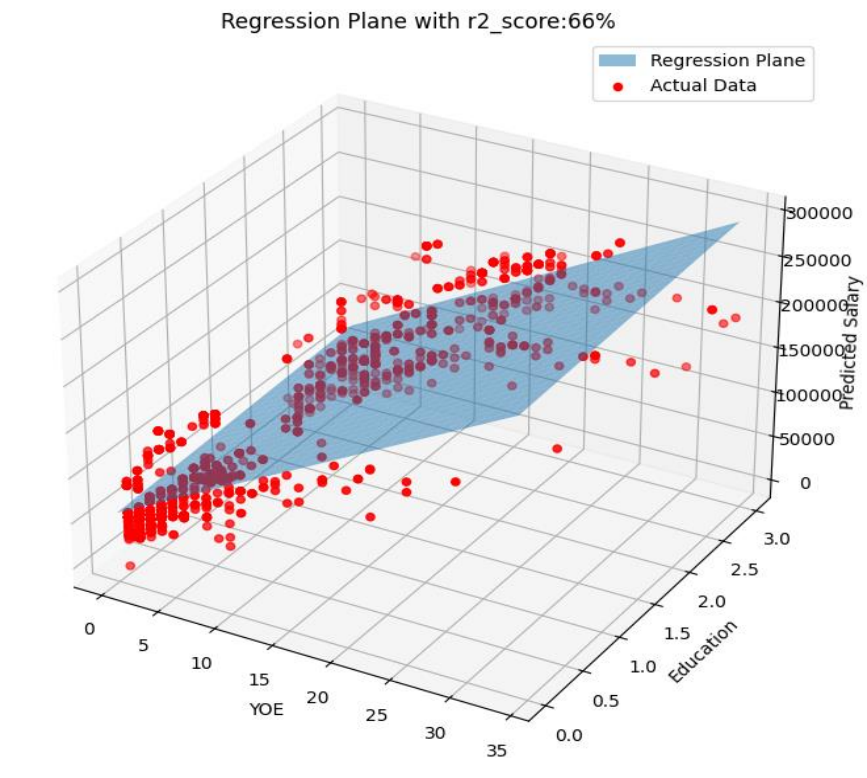
# SALARY BY DEGREE

- It show the relation between variables in heat map
- Years of experience and education_encoded have 50% more relation with price
- This graph helps us to choose columns for regression model

# SALARY VS EXPERIENCE(REGRESSION PLOT)

- There are total three variables in regression expression .
- So a 2d graph can't explain this data

# REGRESSION PLANE ($R^2$ = 66%)




No of variables:

- X1:years of experience(YOE)
- X2:Education
- Y:predicated salary

So a 3d plot much useful to explain the regression plot

# MODEL RESULTS

- The Linear Regression model shows a positive relationship between experience and salary.

- Model Performance:
- $R^2$ Score = 0.66 (66%)
- This means the model explains 66% of salary variation.

# CONCLUSION

- Salary increases with experience and education level.
- Machine learning helps organizations make data-driven salary decisions.
- Future work: add more features like job role, location, and skills.